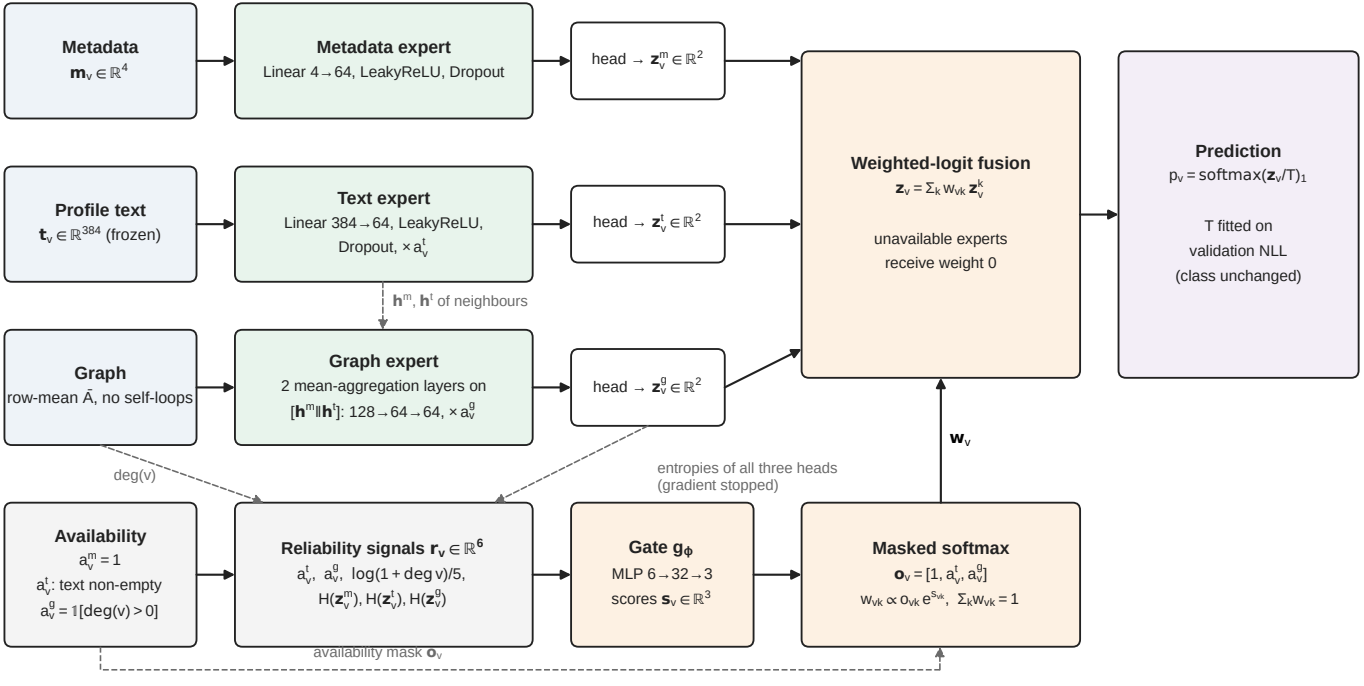
Metadata
$\mathbf{m}_v \in \mathbb{R}^4$
Metadata expert
Linear 4 → 64, LeakyReLU, Dropout
head → $\mathbf{z}_v^m \in \mathbb{R}^2$
Profile text
$\mathbf{t}_v \in \mathbb{R}^{384}$ (frozen)
Text expert
Linear 384 → 64, LeakyReLU, Dropout, × a_v^t
head → $\mathbf{z}_v^t \in \mathbb{R}^2$
$\mathbf{h}^m$, $\mathbf{h}^t$ of neighbours
Graph
row-mean $\bar{A}$, no self-loops
Graph expert
2 mean-aggregation layers on $[\mathbf{h}^m \| \mathbf{h}^t]$: 128 → 64 → 64, × a_v^g
head → $\mathbf{z}_v^g \in \mathbb{R}^2$
Weighted-logit fusion
$\mathbf{z}_v = \Sigma_k w_{vk} \mathbf{z}_v^k$
unavailable experts receive weight 0
Prediction
$p_v = \mathrm{softmax}(\mathbf{z}_v/T)_1$
T fitted on validation NLL (class unchanged)
deg(v)
entropies of all three heads (gradient stopped)
$\mathbf{w}_v$
Availability
$a_v^m = 1$
a_v^t: text non-empty
$a_v^g = \mathbb{1}[\deg(v) > 0]$
Reliability signals $\mathbf{r}_v \in \mathbb{R}^6$
a_v^t, a_v^g, $\log(1 + \deg v)/5$,
$H(\mathbf{z}_v^m)$, $H(\mathbf{z}_v^t)$, $H(\mathbf{z}_v^g)$
Gate g_ϕ
MLP 6 → 32 → 3
scores $\mathbf{s}_v \in \mathbb{R}^3$
Masked softmax
$\mathbf{o}_v = [1, a_v^t, a_v^g]$
$w_{vk} \propto o_{vk} e^{s_{vk}}$, $\Sigma_k w_{vk} = 1$
availability mask $\mathbf{o}_v$